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EV Charging Infrastructure Development Using Machine Learning

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***Abstract*— In the recent years, electric vehicles have emerged as not just an alternative but a replacement for internal combustion engine vehicles. In a few years, most vehicles, both private and commercial will be electric. Considering this, reliable predictions need to be made regarding many different factors, so as to smoothly facilitate the shift from internal combustion engine vehicles to electric vehicles. This paper presents a machine learning model that will predict the possible number of electric vehicles that could arise in localities previously dominated by conventional means of transport, so that the EV charging stations can be installed in the right place. This Paper can be used by anybody who is planning to invest in EV charging infrastructure at any place as well as administrative/governing bodies.**

Keywords—machine learning, charging station, electric vehicle

I. Introduction

A mix of technological advancements in agricultural productivity, sanitation and medical facilities has reduced mortality which has caused an exponential population growth. This population growth resulted in causing climatic change, global warming and pollution. The contribution of humanity towards deterioration of Earth's biosphere is massive. Hence, to judiciously utilize the resources, Sustainable Development needs to be adopted. The United Nations in 2015 adopted seventeen Sustainable Development Goals to direct global efforts for the betterment of Earth and life on it. The main agenda focused on entirely reversing climate change and protecting the planet.[1]

Due to this growth in population the needs of the people including transportation quality is increasing. More and more people prefer private means of transportation for their daily chores. As a result there is a significant increase in the number of vehicles being manufactured. Taking into account the environmental issues associated with more number of vehicles being manufactured like climate change, global warming, etc. there is an urgency to switch to using Electric Vehicles (EVs). Continuous advancements in technology allow us to harness clean and renewable means of energy which can power the EVs. Use of clean energy will have a great positive impact on the environment as it will help it mitigating most of the environmental issues caused by burning of fossil fuels. The United States Environment Protection Agency (EPA) in 2019 in their annual report "The Inventory of Greenhouse Gas Emission and Sinks" estimated that twenty-nine percent of greenhouse gas emissions came from the transportation sector caused by burning of fossil fuels by cars, trains, ships etc.

The conversion of IC engine vehicles into EVs is the key solution to reducing emission of greenhouse gases. But there are many challenges for this transformation to get implemented successfully. Some of the hindrances include popularity, range of travel, charging infrastructure and reliability. EVs are gaining popularity recently due to constant efforts put forward by nominal number people and companies. Range of travel and other peripherals provided by EVs depends on the performance of the battery used in the specific EV. The integral part of an EV is its battery and its management system. Battery motor technology is a field with constant influx of innovation which is a driving factor for boosting up the notion of EVs being the best amongst the people. Charging infrastructure is one of the most challenging factors in the successful advent of EVs. Efforts are being made to establish more electric charging infrastructure in the n ear future, at appropriate places where there is necessity, usability and profitability. The hindrance which arises in opening electric charging infrastructures is that they are less likely to yield high profits. Whereas in case of opening a gas station, profitability is assured as the high number and density of IC engine vehicles that needs to be refuelled.[2]

Installing EV charging station seems to be a more risky business due to the less number of electric vehicles prevalent now.[3] [4] Also, low number of EVs means low requirement of recharging and as a result ,installing more EV charging stations decreases profitability. In addition to that, installing an EV charging station is a cumbersome task. Hence in order to minimize the risk of installing an EV charging station a more calculated decision about where to set up the recharging station should be adopted rather than establishing recharging stations randomly on a vague basis. Machine Learning models can be developed using different algorithms to predict the possible number of electric vehicles that could arise in an area previously dominated by conventional means of transport. Also, these ML algorithms can be employed in finding out the regions where an electric charging station must be installed.[5]

The research gap found is that there is no method of finding out the relationship between the number of electric vehicles present in an area and the number of electric charging stations that needs to be installed. In this hour of the technological era, more advancements are being carried out in the sector of electric vehicles as they are very eco-friendly and hence it is the need of the hour to propose such a model so that everyone can benefit out of it.

II. CHARGING INFRASTRUCTURE ISSUES

Since electric vehicles are a superior alternative to internal combustion engine vehicles for our environmental aspects, there are few downsides to these electric vehicles. One of the main issues with an electric vehicle is its charging time and travel range. Most of the EVs have a range of 300-400 Km in a single charge and have a charging time of more than 1hr 30min, even using a fast charger. Due to this, the charging stations should be able to charge multiple vehicles for a long time. [6]

The other major problem regarding charging station is to determine the right place to set up a charging station. As electric vehicles are not as common as internal combustion engine vehicles, it won't be as profitable as opening an oil pump and is also fairly risky. For setting up charging infrastructure there should be the availability of abundant amount of power and multiple charging spots which can charge electric vehicles for a long time.[7] With each coming day, the speed charging technology is increasing so a charging infrastructure should be able to keep up with providing new technology for the users.

Mainly, in the place where charging station is being set up, there should be a lot of electric vehicle users present. Otherwise opening a charging station will be a big mistake. That is where this paper comes into handy. The model developed in this paper will tell what places will be best suited for opening a charging station. Thus, people can open a charging infrastructure without taking much risk and make it a profitable business.

III. Machine Learning Models for Charging station Infrastructure

Two models have been presented in this paper to solve this problem. Model 1 uses linear regression method, which puts forward a machine learning model to predict the number of electric vehicles running in Pune, and model 2 which is finds the best regression model for predicting the probability of installing a charging station.[8] [9]

A. Linear Regression method

There are two types of regression to solve this problem. First method of regression is linear regression and the second one is polynomial regression. Linear regression is a supervised machine learning technique with a continuous and constant slope projected output. It is used to predict values within a continuous range rather than aiming to classify data into categories, (e.g. sales, price). For minimization, least squares cost function is used. Polynomial regression has a higher order variable in it and this is due to the relationship between the independent and dependent variables does not have to be linear, therefore it has more flexibility in the datasets and circumstances to work with. When simple linear regression is not sufficient for the analysis of data, this approach can be used.

Some input/learning parameters that are needed for writing a suitable machine learning code to find the optimal place for charging infrastructure are:

1) Population of each ward in the city
2) Total number of vehicles in the ward
3) Average income of a resident in the ward
4) Average distance traveled by car in the ward

This data will be collected in a spreadsheet and will be saved in a csv file which can be accessed by the python codes for the machine learning purposes.

The next step is to train the model to finding the minimum degree of the regression. Numpy, pandas and matplotlib libraries should be imported into the code then data set should be imported hence introducing and giving the data stored in the spreadsheet to the variables x and y.

After that the model is trained and minimum degree and mean squared error (MSE) is found by using the algorithm below.

Step 1 : Create a list named error_Number_of_degree
Step 2 : Initialise the value of i as 1 and repeat the steps from 3 to 9 while value of i of less than 11
Step 3 : Import libraries such as PolynomialFeatures and linear_model
Step 4 : Set the value of poly_reg, x_poly, x_ and y as PolynomialFeatures(degree=i) ,polyreg.fit_transform(x),polyreg.fit_ transform(x) and y.flatten() respectively.
Step 5: Set value of clf as linear_model.LinearRegression()
Step 6 : Call the variables in clf.fit()
Step 7 : Set y_pred as clf.predict(x_)
Step 8 : Import mean_squared_error library
Step 9 : Set value of y_true and error as y, mean_squared_error(y_true,y_pred) respectively. Then add the each value of error to error_Number _of_degree list
Step 10 : clf.coef

After that a graph of MSE is prepared taking degree as the x-axis and MSE as the y-axis using the help of plot function in the library matplotlib. This graph is shown in the Figure 1.

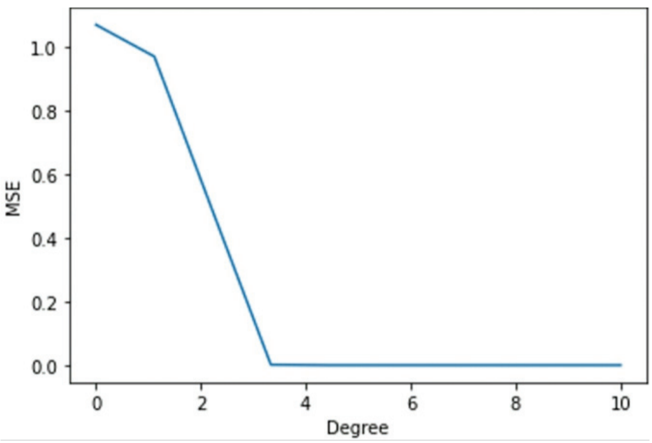

Fig. 1. Graph between MSE and Degree in Model 1

From the Figure 1 it can be understood that the minimum polynomial degree 3 has the lowest MSE. But 3 cannot be taken as it over fits the data.

Now, the covariance between the different input data set is to be found. This is made possible by the help of covariance_ function which is defined in the library Empirical Covariance.

The covariance provides a natural measure of the association between two variables, and it appears in the analysis of many problems in quantitative genetics including the resemblance between relatives, the correlation between characters, and measures of selection.

B. *Finding Best Regression model*

In this model, the best regression model for predicting the probability of installing a charging station for that data is done.[10] [11] The input data that need to be collected to store in the spreadsheet are:

1) Population of the ward
2) Total number of vehicles in the ward
3) Average distance travelled by the car in the ward.

The probability of installing a charging station depicted in a heat map is the output of this model.

The next step is to code the program in a python IDE for that first library files such as numpy, pandas and matplotlib should be introduced into the program. Importing the dataset into the program will be the next step, for that the values stored in the csv files are read and the values are accessed by the variables x and y respectively. Next, the model with the least MSE should be found. The following algorithm should be followed.

Step 1 : Initialize the value of i as 1 and repeat the steps till step from 2 to 11 until i is less than 10
Step 2 : import libraries such as PolynomialFeatures linear_model
Step 3 : Assign the value of poly_reg as PolynomialFeatures(degree = i) and that of x_poly and x_ as poly_reg.fit_transform(x)
Step 4 : Now convert the y array into a single dimension using the flatten( ).
Step 5 : Now assign the value of clf as linear_model.LinearRegression( )
Step 6 : Now pass the x_ and y into clf
Step 7 : y_ pred is assigned the value of clf.predict(x_)
Step 8 : Now import mean_squared_error library
Step 9 : y_true = y
Step 10 : Now assign the value of error as MSE of y_true, y_pred

Now create the graph to visualize MSE for each degree, this is made possible by the help of plt.plot( ) function. This function plots the graph between MSE and each degree as per the given instruction and this graph is shown in Figure 2.

It can be concluded that the model with degree 6 has the best fit from the graph. Then the covariance between different input variables will be carried out by the same approach as in model 1. Thus the covariance between the different input data set is found.

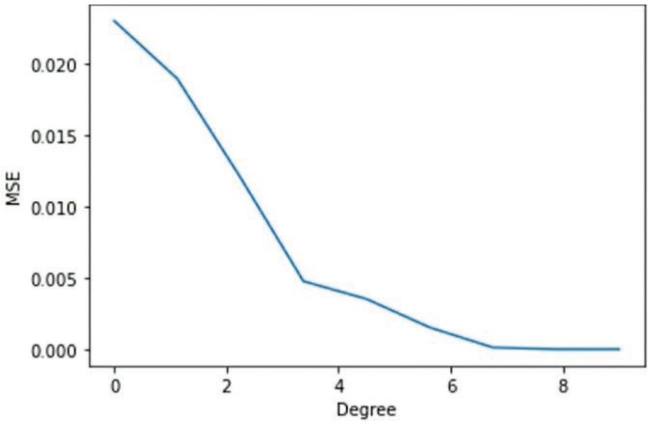

Fig. 2. Graph between MSE and Degree in Model 2

C. *Locating electric vehicle charging station*

The regression model fails to provide us with accurate results when it is fed with the data. Because higher powered variables are used. Thus, in order to increase the output accuracy, classification data must be done so that data will be obtained in the form of different categories. Random forests are used for this purpose.[12] Random forests or random decision forests, are an ensemble learning method for regression, classification and other problems that works by training a large number of decision trees. Majority of trees during classification tasks use the output class of random forest whereas the output of regression task is the mean or average prediction of an individual tree.[13] The problem of decision trees over fitting their training set is addressed by random decision forests.

In this model, different wards or areas in the Pune city are classified according to the probability of having success or yielding profits when a charging station will be installed. The different areas are classified into four categories ranging from best to worst. Even though this paper is mainly focused on Pune city, this paper also gives algorithm for any general situation, so it adaptable. For implementing our model random forest, bagging is used. [14] The implementation is done by bagging. It uses a simple approach that shows up in statistical analysis again and again — improving the estimate of one by combining the estimates of many. Bagging constructs 'n' classification trees using bootstrap sampling of the training data and then combines their predictions to produce a final meta-prediction. Bagging is done by starting by creating an empty array named bagged model accuracy followed by defining a second array namely test split data with values fed into it. The next step is to include all the libraries such as pandas, metrics, DecisionTreeClassifier, train test split along with the dataset which contains all the data required with a .csv extension. Now the data is pre-processed by declaring the input and output parameters subsequently. As our dataset comprises of multiple classes or categories of data multi-class perceptron algorithm is used for classifying multiple classes of data. Along with the algorithm regularization, parameter is added and a lambda parameter is included in the cost function in order to prevent the over fitting of data.

For evaluating the machine learning model train test split is imported from sklearn.model selection. RandomForestClassifier is also imported as our model is a classification based. Now the model is categorized into five parts namely 'Trial model 1', 'Trial model 2', 'Trial model

3', 'Trial model 4' and 'Trial model 5'. In each models the test size of the data set is varied and the corresponding accuracy on training as well as testing data set is recorded. All the five models are then tested on 10% data followed by bagging. Increasing stability of models by improving accuracy, reducing variance and eliminating over fitting during decision trees is done by bagging.

Finally the accuracy of all the five models and the accuracy of bagged Random forest is displayed using metrics.balanced_accuracy_score() function. test_split_data vs bagged_model_accuracy is plotted below in Figure 3.

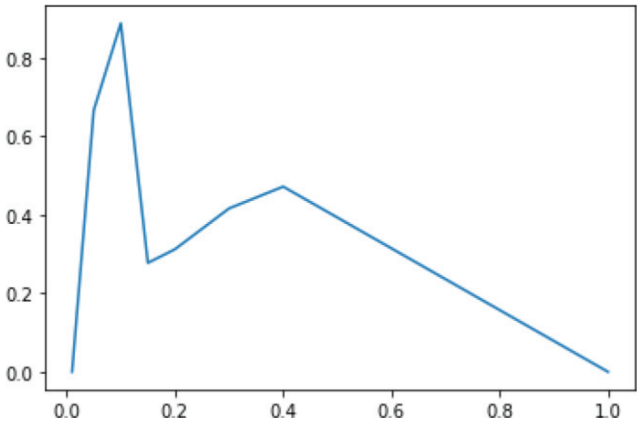

Fig. 3. Graph showing relation between test_spit_data and bagged_model_accuracy

From the above Figure 3 it's been concluded that if our test data is split as 10% the best output is obtained.

D. Clustering the data

Clustering or cluster analysis is a machine learning technique, which groups the un-labelled dataset. It is a way of grouping the data points into different clusters, consisting of similar data points. The objects with the possible similarities remain in a group that has less or no similarities with another group. K-Means Clustering is an Unsupervised Learning algorithm, it groups the unlabeled dataset into different clusters. Here K defines the number of pre-defined clusters that need to be created in the process, for K=2, there are two clusters, and for K=3, there are three clusters, and so on. In this section the similar data are clustered together which help in understanding and manipulating the data easily. [15] - [18]

Python is used to write the program for clustering. In this program, firstly all the libraries like matplotlib, numpy, KMeans and pandas are imported. Then all the data which are stored in the .csv extension is accessed with the help of pd.read_csv sunction and the values are taken with help of x and y variables. Elbow method is then used to find the optimum number of clusters.

In the Elbow method, the number of clusters (K) are varied from 1 – 10. For each value of K, WCSS (Within-Cluster Sum of Square) is calculated. WCSS is the sum of squared distance between each point and the centroid in a cluster. When the WCSS with the K value is plotted, the plot looks like an Elbow shape. As the number of clusters increases, the WCSS value will start to decrease. WCSS value is largest when K = 1. While analyzing the figure 4 it can be seen that the graph will rapidly change at a point and thus creating an elbow shape. From this point, the graph starts to move almost parallel to the X-axis. The K value corresponding to this point is the optimal K value or an optimal number of clusters.

Hence using this method a graph is plotted between WCSS and number of clusters as shown in Fig 4 using the help of plt.plot function. From the Graph show in Figure 4 it can be concluded that the optimum number of clusters should be 5 since the WCSS value decreases slowly from that point onwards.

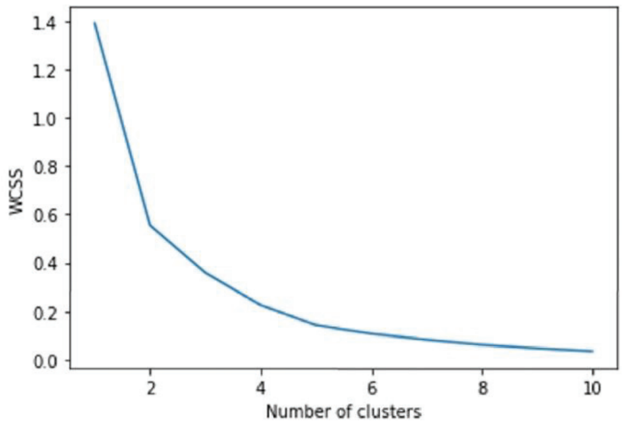

Fig. 4. Graph of the elbow method

Next step is to train the K-Means model on the dataset stored in the .csv extension. For that fit_predict() function is used and draw the scatter plot of the clusters for x1 and x2, x1 and x3 and for x2 and x3 using the help of plt.scatter function.[19] The result of these program to draw scatter plot is shown in Figure 5, Figure 6 and Figure 7 respectively.

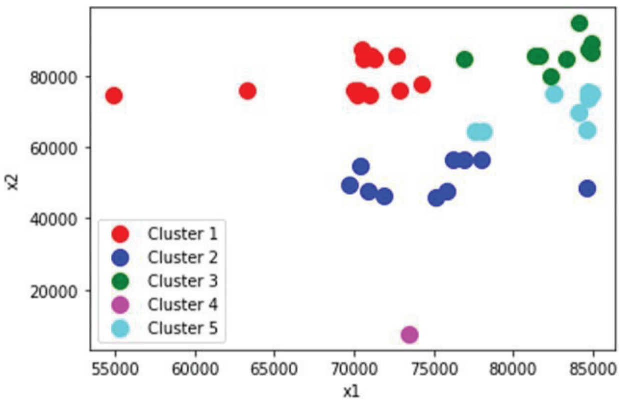

Fig. 5. Scatter plot of Clusters for x1 and x2

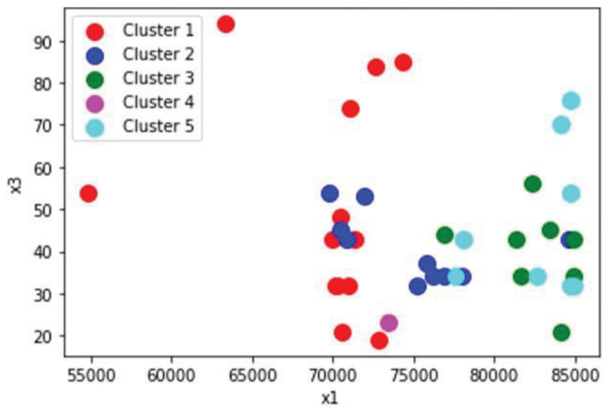

Fig. 6. Scatter plot of Clusters for x1 and x3

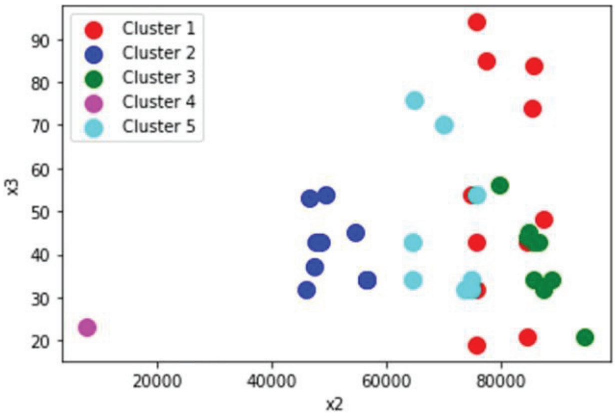

Fig. 7. Scatter plot for x2 and x3

After passing the data that has been collected through this model, it will give the output which says the best place to open a charging station in Pune city. A heat map is used to depict this information given below in Fig 8. [20] In this heat map it says that the white area is the place that the data is not available, the rose area has 0% - 25% for a success, the red area has a success rate of 25% - 50% and the maroon area has a success rate of 50% - 75% in opening a charging station. Opening a charging station in the maroon area will be a lot more profitable and less risky than opening a charging station in rose area.

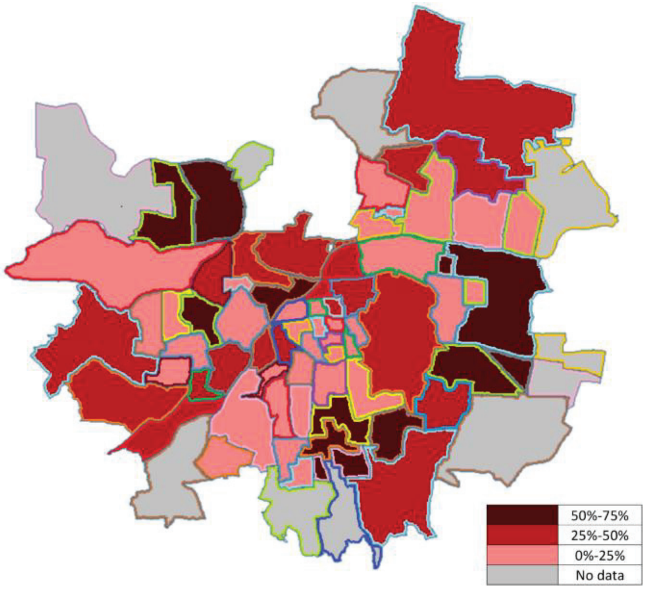

Fig. 8. Heat map depicting the probability of opening a charging station in Pune city

IV. CONCLUSIONS

This paper comprises of a machine learning model which is used to predict the increase in the number of electric vehicles in a particular area which was previously dominated by conventional fossil fuel vehicles, this machine learning model will help us understand the demand for EV charging stations that would arise in the future for a particular area so that EV charging stations could be installed in the right places at the right time. Already, the demand for EVs is on the rise and not going down anytime soon. This paper contributes towards the global issue of pollution and global warming and if done rightly, EV charging stations at the right places could be a major milestone in promoting the use of EVs. Population of the ward, total number of vehicles in that ward, average income of the residents living there and the average distance travelled by a car in each ward are the variables that are chosen in this study. This model will greatly help in scaling the businesses of EV companies and its subsidiaries as they can figure out the right place where demand of EV vehicles is high and subsequently set up their EV charging stations. Also, this shall further motivate the public to adopt EVs as a preferred means of transport over conventional transport.

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